

What Drives Firm-Level Stock Returns?

Teaching note

Vuolteenaho (2002), *Journal of Finance*

ACCTG 597A · Fall 2026 · Sam Bonsall · Penn State

The aggregate literature had an answer; the firm level had none

By identity, an unexpected stock return must reflect news about future cash flows, news about future discount rates, or both. The aggregate evidence said expected-return news dominates.

"There is a substantial body of research measuring the relative importance of cash-flow and expected-return news for aggregate portfolio returns (e.g., Campbell (1991) and Campbell and Ammer (1993)), but virtually no evidence on the relative importance of these components at the firm level."

Vuolteenaho (2002, p. 233)

The module examines what had to change for the aggregate technology to run on 36,791 firm-years, and what the transplant found.

The transplant required an accounting substitution

The Campbell-Shiller (1988a) identity expresses the log dividend-price ratio as a discounted sum of future dividend growth and future returns. At the firm level the dividend primitive fails: dividend policies are unstable, heterogeneous, and often zero (p. 234, fn. 1).

Vuolteenaho's substitution: impose clean surplus and let ROE replace dividend growth, so dividends drop out of the identity entirely (eqs. 1-2, p. 235):

$$B_t = B_{t-1} + X_t - D_t \qquad \theta_{t-1} = k_{t-1} + \sum_{j=0}^{\infty} \rho^j r_{t+j} - \sum_{j=0}^{\infty} \rho^j (e_{t+j} - f_{t+j})$$

The log book-to-market ratio θ is low when expected future excess ROEs ($e - f$) are high or expected future excess returns r are low; $\rho = 0.967$ in-sample.

The identity rests on the clean-surplus relation, yet the reference list contains no accounting valuation paper; the connection to Ohlson (1995) returns on the citation slide.

Differencing the identity yields a news decomposition

Take the change in expectations of the identity from $t - 1$ to t and reorganize (eqs. 3-4, p. 236):

$$r_t - E_{t-1}r_t = N_{cf,t} - N_{r,t}$$
$$N_{cf,t} \equiv \Delta E_t \sum_{j=0}^{\infty} \rho^j (e_{t+j} - f_{t+j}) + \kappa_t, \quad N_{r,t} \equiv \Delta E_t \sum_{j=1}^{\infty} \rho^j r_{t+j}$$

The variance decomposition follows (eq. 5):

$$\text{var}(r_t - E_{t-1}r_t) = \text{var}(N_{r,t}) + \text{var}(N_{cf,t}) - 2 \text{cov}(N_{r,t}, N_{cf,t}).$$

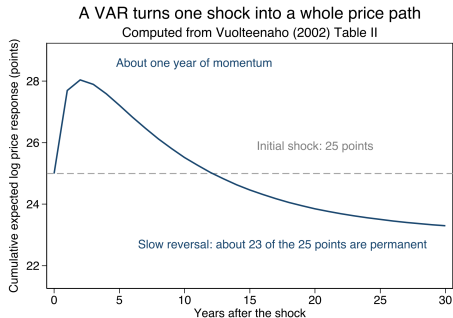
One observation carries the research design: the news terms are revisions in infinite-horizon expectations, which no one observes. The paper's empirical content therefore depends entirely on a model of how expectations are formed. That model is the VAR.

The VAR supplies the expectations the identity needs

State vector z_t : market-adjusted log return, log book-to-market, log profitability; $z_{i,t} = \Gamma z_{i,t-1} + u_{i,t}$ (eq. 6, p. 240).

Each variable is regressed on last year's three; Γ collects the coefficients; u_t is today's surprise. The forecast j years ahead is $\Gamma^j z_t$, so one shock revises the whole expected path.

The state variables span the documented return-predictability anomalies (p. 234).



One news term is modeled; the other is a residual

The discounted infinite-horizon forecast revision is a geometric sum in Γ , the VAR analog of a perpetuity formula (eqs. 7-8, p. 240):

$$\lambda' \equiv e1' \rho \Gamma (I - \rho \Gamma)^{-1}, \quad N_{r,t} = \lambda' u_{i,t}, \quad N_{cf,t} = (e1' + \lambda') u_{i,t}$$

The three variance-decomposition numbers follow from Γ and the error covariance Σ alone: $\text{var}(N_r) = \lambda' \Sigma \lambda$, $\text{var}(N_{cf}) = (e1' + \lambda') \Sigma (e1 + \lambda)$, $\text{cov} = \lambda' \Sigma (e1 + \lambda)$.

Two design observations. First, if returns are unpredictable (the first row of Γ is zero), expected-return news is identically zero and every return is cash-flow news by construction. Second, expected-return news is computed directly from the VAR, while cash-flow news is backed out as the residual that makes the identity hold; the consequences of that asymmetry return two slides from the end.

The design validates its inputs before using them

Sample. The 1954-1996 CRSP-COMPUSTAT intersection, 36,791 firm-years, annual frequency. No data requirements are imposed on year- t variables, so selection operates only through lagged state variables (p. 237). Each firm is redefined as 90 percent stock and 10 percent Treasury bills to keep log transforms well behaved (p. 238).

Estimation. Pooled WLS, one prediction regression per state variable, weighting each annual cross-section equally; Rogers (1983, 1993) cluster-robust and Shao-Rao (1993) delete-one-cross-section jackknife standard errors, years before clustering became the accounting default. Ignoring cross-sectional correlation “can yield standard errors that are biased downwards by a factor of five” (Appendix B, p. 260).

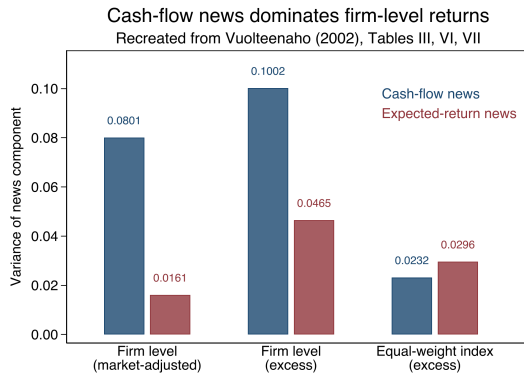
Identity validation. The approximation is tested, not assumed: Table AI (p. 261) yields $\rho = 0.967$ with $R^2 = 99.82$ percent under clean-surplus ROE, falling to 88.25 percent under GAAP ROE, a quantified measure of how far GAAP departs from clean surplus.

Result 1: at the firm level, cash-flow news dominates

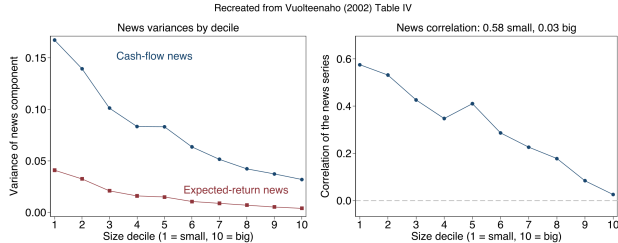
Market-adjusted returns (Table III): cash-flow-news variance 0.0801 against 0.0161, roughly five to one; expected-return news is about one-quarter of total unexpected-return variance.

Excess returns (Table VI): 0.1002 against 0.0465, roughly two to one.

The introduction prints 0.0645 for the excess expected-return-news variance; the table figure 0.0465 matches the stated 22 percent standard deviation.



Result 2: the decomposition is a function of firm size

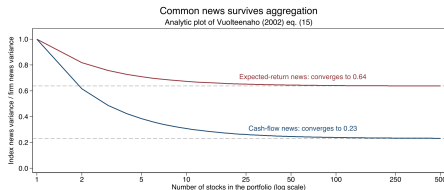


For the largest decile, cash-flow-news variance is eight times expected-return-news variance and the news correlation is near zero; for the smallest, the correlation is 0.58 (Table IV). The return response to cash-flow news rises from $b = 0.72$ (small) to 0.99 (large). Two interpretations: underreaction to small-stock cash-flow news, or zero-NPV investment, in which a newly announced high-risk project must promise commensurately high expected cash flows, generating the correlation without mispricing (pp. 248-250).

Result 3: aggregation reverses the answer

"It appears that while cash-flow information is largely firm specific (such as success or failure of a specific product), expected-return information is predominantly driven by systematic, macroeconomic components."

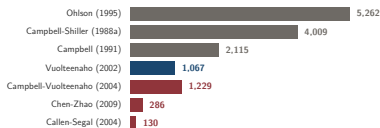
Vuolteenaho (2002, p. 253)



Averaging the fitted news series into an equal-weight index restores the Campbell-Ammer (1993) ranking: expected-return-news variance 0.0296, cash-flow-news variance 0.0232 (Table VII).

A residual-based decomposition inherits every omitted state variable

Chen and Zhao (2009) show the variance shares are sensitive to the state-variable list and that reversing which news term is the residual can reverse the conclusion; their Treasury-bond illustration attributes substantial cash-flow news to securities that have none. Vuolteenaho's Table VIII is the partial defense: computed directly from clean-surplus ROEs, the approximation error is small and the residual method conservative.



Citations (OpenAlex, Aug. 2026); journal records, except Campbell (1991), which combines its split records; gray = technology and the uncited neighbor, navy = focal, maroon = descendants and critique

Of the 1,069 indexed works citing Vuolteenaho (2002), only 98 (roughly nine percent) also cite the critique; the Callen-Segal (2010) primer's indexed reference list, compiled the year after Chen-Zhao, does not include it.

The technology crossed into accounting

The adoption trail. Callen-Segal (2004) disaggregate cash-flow news into accrual and operating components, restoring the Ohlson (1995) citation the original omitted; Callen-Hope-Segal (2005) split earnings news into domestic and foreign components; Easton-Monahan (2005) use the news terms to referee the implied-cost-of-capital literature; Callen (2009) supplies the theory; Callen-Segal (2010) codify the recipe in a primer.

Candidate designs going forward. Any question of the form *how much of the market response reflects revised cash-flow expectations versus revised discount rates* fits the machinery: disclosure and reporting-quality events, ESG and climate news, credit-market events through an analogous bond identity, and direct news measures replacing the VAR residual (the Chen-Da-Zhao 2013 route).

The critique implies a discipline for all of them: state and justify the state variables; compute the accounting-side news directly when clean surplus permits.

Summary

- Vuolteenaho (2002) transplants the Campbell (1991) aggregate variance decomposition to the firm level by substituting clean-surplus ROE for dividend growth, and finds firm-level returns predominantly driven by cash-flow news, the reverse of the aggregate ranking; the two reconcile because cash-flow news diversifies away in portfolios while expected-return news does not.
- The design illustrates creative technology transfer: the barrier was an accounting problem, unmodelable dividend policy, and the solution was an accounting identity, imported without citation to the accounting literature that had formalized it.
- The decomposition also carries a durable research-design lesson: a news term constructed as a residual absorbs every omitted state variable, and the citation record suggests the literature adopted the technology considerably more thoroughly than it adopted the critique.

Vuolteenaho (2002) · Campbell (1991) · Callen-Segal (2004) · Chen-Zhao (2009) ·
github.com/sbb151/acctg597a-research-design